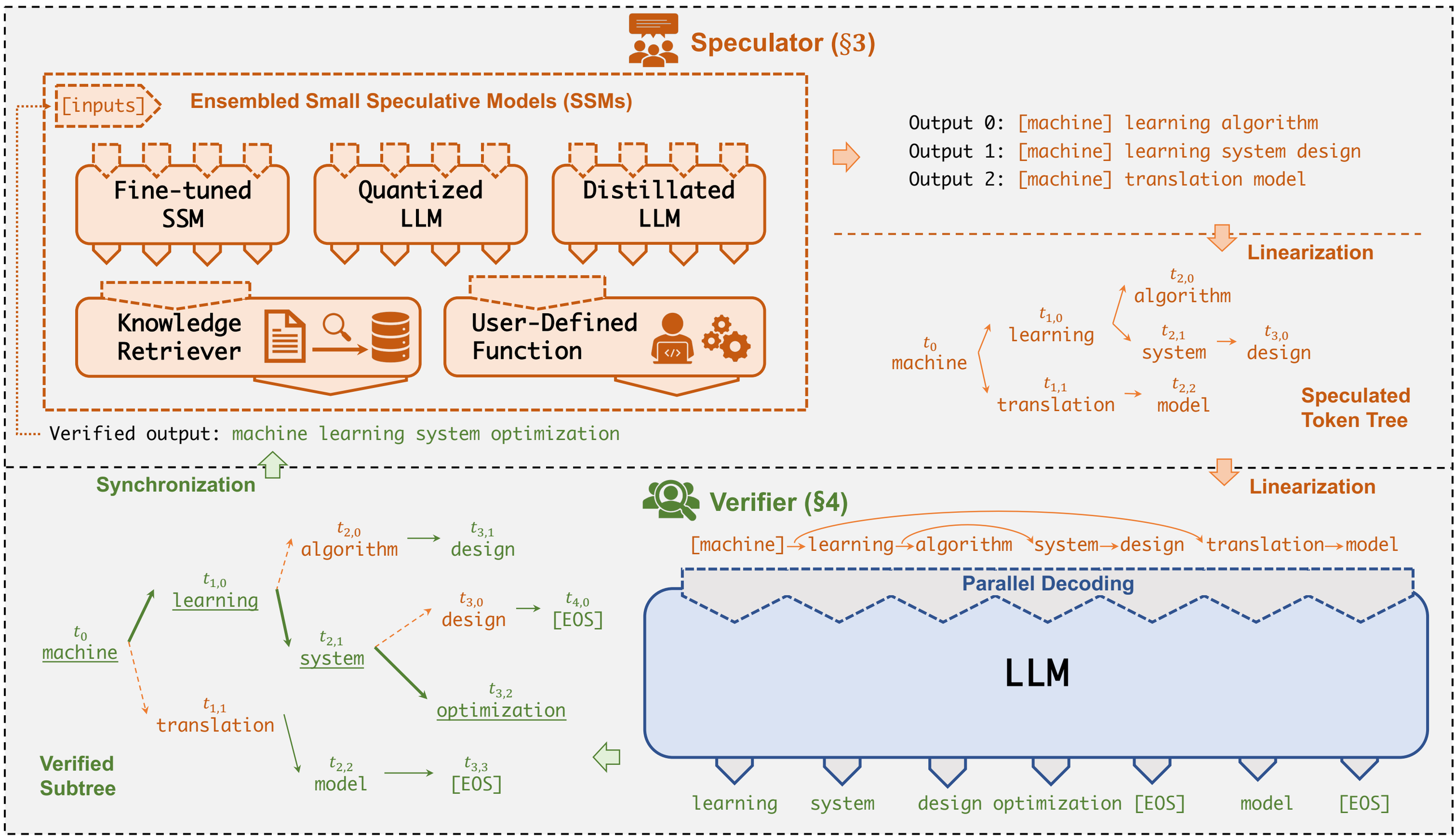
Speculator (§3)
[inputs]
Ensembled Small Speculative Models (SSMs)
Fine-tuned SSM
Quantized LLM
Distillated LLM
Knowledge Retriever
User-Defined Function
Output 0: [machine] learning algorithm
Output 1: [machine] learning system design
Output 2: [machine] translation model
Linearization
t_0 machine
$t_{1,0}$ learning
$t_{2,0}$ algorithm
$t_{2,1}$ system
$t_{3,0}$ design
$t_{1,1}$ translation
$t_{2,2}$ model
Speculated Token Tree
Verified output: machine learning system optimization
Synchronization
Verified Subtree
$t_{2,0}$ algorithm
$t_{3,1}$ design
$t_{1,0}$ learning
t_0 machine
$t_{3,0}$ design
$t_{4,0}$ [EOS]
$t_{2,1}$ system
$t_{3,2}$ optimization
$t_{1,1}$ translation
$t_{2,2}$ model
$t_{3,3}$ [EOS]
Verifier (§4)
Linearization
[machine] learning algorithm system design translation model
Parallel Decoding
LLM
learning system design optimization [EOS] model [EOS]